



Research Article

INT8 Activation Ternary or Binary Weights Networks: Unifying Between INT8 and Lower-bit Width Quantization

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ABSTRACT

This paper proposes ternary or binary weights with 8-bit integer activation convolutional neural networks. Our proposed model serves as the middle ground between 8-bit integer and lower than 8-bit precision quantized models. Our empirical experiments established that the conventional 1-bit or 2-bit only-weight quantization methods (i.e., BinaryConnect and ternary weights network) can be used jointly with the 8-bit integer activation quantization. We evaluate our model with the VGG16-like model to operate with the CIFAR10 and CIFAR100 datasets. Our models show competitive results to the general 32-bit floating point model.

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1. Introduction

The invention of residual networks [1] has drastically improved the performances of neural networks (NNs); these networks now far exceed human performances, especially in image recognition tasks. However, NNs require numerous parameters to perform well. When the number of parameters increases, more computational operations are required. This might prove to be challenging when deploying these models to edge devices that lack high computational capacities.

These problems have led to the emergence of several research fields. One such field is quantization, which converts the conventional 32-bit floating-point (FP32) datatype into simpler datatypes, such as the fixed-point datatype and integer datatype. Currently, the default quantization technique supported by major deep learning frameworks (i.e., PyTorch [2] and TensorFlow [3]) is the

8-bit integer (INT8) quantization [4]. INT8 quantization has become the standard because INT8 datatypes and operations are widely supported by vendors of central processing units and graphics processing units. Therefore, the INT8 quantized model can be used across devices. In general, INT8 quantization transforms almost all FP32 parameters to INT8 via an affine transformation. This conversion allows the quantized model to operate primarily with INT8 operations, which are faster than FP32 operations. For example, in the NVIDIA Ampere architecture, NVIDIA A100 delivers approximately 32 times more INT8 operations per second compared with FP32 operations [5]. Furthermore, INT8 reduces the NN's overall memory storage by a factor of four compared with FP32.

INT8 quantization also positively affects FPGA implementations. For example, when we constructed FP32 and INT8 accumulation between two variables with Xilinx Vitis HLS 2021.2 using the target board Xilinx

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